SNOWPRO CORE

EXAM Study Guide

Jan 2025

Agenda

- SnowPro Certification
- SnowPro Core Certification
- SnowPro Core Exam Domains
- Summary for Each Domain
- Sharing by Peers

SNOWPRO Certification

SNOWPRO Certification

Total 6 SnowPro Certifications are available

SnowPro Core

SnowPro Advanced Architect

SnowPro Advanced Administrator

SnowPro Advanced Data Engineer

SnowPro Advanced Data Analyst

SnowPro Advanced Data Scientist

SNOWPRO CORE Certification

SNOWPRO CORE Certification

Snow Pro Core verifies Implementation and Migration Skills and Knowledge for Snowflake

Required Knowledge

- Data Loading and Transformation in Snowflake
- Virtual Warehouse Performance and Concurrency
- DDL and DML Queries
- Using Semi-Structured and Unstructured Data
- Cloning and Time Travel
- Data Sharing
- Snowflake Account Structure and Management

Exam Candidates

- Solutions Architect
- Data Engineer
- Snowflake Account Administrator
- Database Administrator
- Data Scientist
- Data Analyst
- Application Developer

Outline for SNOWPRO CORE Exam

SNOWPRO CORE CERTIFICATION EXAM

Total Number of Questions: 100

Question Types: Multiple Select, Multiple Choice

Time Limit: 115 minutes

Languages: English

Registration Fee: $175 USD

Passing Score: 750 + Scaled Scoring from 0 - 1000

Delivery Options:

- Online Proctoring
- Onsite Testing Centers

Click here for information on scheduling your exam.

Weights for each domain for SNOWPRO Exam

Domain	Domain Weightings
1.0 Snowflake AI Data Cloud Features and Architecture	24%
2.0 Account Access and Security	18%
3.0 Performance and Cost Optimization Concepts	16%
4.0 Data Loading and Unloading	12%
5.0 Data Transformations	18%
6.0 Data Protection and Data Sharing	12%

Outline for each domain

1. Snowflake Data Cloud Features and Architecture

1.1 Outline key features of the Snowflake Data Cloud.

- Elastic Storage
- Elastic Compute
- Snowflake's three distinct layers
- Data Cloud/ Data Exchange/ Partner Network
- Cloud partner categories

1.2 Outline key Snowflake tools and user interfaces.

- Snowflake User Interfaces (UI)
- Snowsight
- Snowflake connectors
- Snowflake drivers
- SQL scripting
- Snowpark

1.3 Outline Snowflake's catalog and objects.

- Databases
- Schemas
- Tables Types
- View Types
- Data types
- User-Defined Functions (UDFs) and User Defined Table Functions (UDTFs)
- Stored Procedures
- Streams
- Tasks
- Pipes
- Shares
- Sequences

1.4 Outline Snowflake storage concepts.

- Micro partitions
- Types of column metadata clustering
- Data storage monitoring
- Search optimization service

Outline for each domain

2. Account Access and Security

2.1 Outline security principles.

- Network security and policies
- Multi-Factor Authentication (MFA)
- Federated authentication
- Single Sign-On (SSO)

2.2 Define the entities and roles that are used in Snowflake.

- Outline how privileges can be granted and revoked
- Explain role hierarchy and privilege inheritance

2.3 Outline data governance capabilities in Snowflake.

- Accounts
- Organizations
- Databases
- Secure views
- Information schemas
- Access history and read support

Outline for each domain

3. Performance Concepts

3.1 Explain the use of the Query Profile.

- Explain plans
- Data spilling
- Use of the data cache
- Micro-partition pruning
- Query history

3.2 Explain virtual warehouse configurations.

- Multi-clustering
- Warehouse sizing
- Warehouse settings and access

3.3 Outline virtual warehouse performance tools.

- Monitoring warehouse loads
- Query performance
- Scaling up compared to scaling out
- Resource monitors

3.4 Optimize query performance.

- Describe the use of materialized views
- Use of specific SELECT commands

Outline for each domain

4. Data Loading and Unloading Study Resources

4.1 Define concepts and best practices that should be considered when loading data.

- Stages and stage types
- File size
- File formats
- Folder structures
- Adhoc/bulk loading using the Snowflake UI

4.2 Outline different commands used to load data and when they should be used.

- CREATE PIPE
- COPY INTO
- GET
- INSERT/INSERT OVERWRITE
- PUT
- STREAM
- TASK
- VALIDATE

4.3 Define concepts and best practices that should be considered when unloading data.

- File formats
- Empty strings and NULL values
- Unloading to a single file
- Unloading relational tables

4.4 Outline the different commands used to unload data and when they should be used.

- LIST
- COPY INTO
- CREATE FILE FORMAT
- CREATE FILE FORMAT … CLONE
- ALTER FILE FORMAT
- DROP FILE FORMAT
- DESCRIBE FILE FORMAT
- SHOW FILE FORMAT

Outline for each domain

5. Data Transformations

5.1 Explain how to work with standard data.

- Estimating functions
- Sampling
- Supported function types
- User-Defined Functions (UDFs) and stored procedures

5.2 Explain how to work with semi-structured data.

- Supported file formats, data types, and sizes
- VARIANT column
- Flattening the nested structure

5.3 Explain how to work with unstructured data.

- Define and use directory tables
- SQL file functions
- Outline the purpose of User-Defined Functions (UDFs) for data analysis

Outline for each domain

6. Data Protection and Data Sharing

6.1 Outline Continuous Data Protection with Snowflake.

- Time Travel
- Fail-safe
- Data Encryption
- Cloning
- Replication

6.2 Outline Snowflake data sharing capabilities.

- Account types
- Snowflake Marketplace and Data Exchange
- Private data exchange
- Access control options
- Shares

Quiz

1. **Which type of data integration tools leverage Snowflake's scalable compute for data transformation?**

   A. Database replication

   B. ELT

   C. ETL

   D. Streaming

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2. What is the maximum number of consumer accounts that can be added to a share object?

A. 1

B. 10

C. 100

D. Unlimited

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A. Pruning

B. Indexing

C. Map Reduce

D. B-Tree

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A. Auto-suspend

B. Storage size

C. Auto-resume

D. Cache size

E. Default role

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   B. Storage size

   C. **Auto-resume**

   D. Cache size

   E. Default role

5. Which role in Snowflake allows a user to administer users and manage all database objects?

A. SYSADMIN

B. SECURITYADMIN

C. ACCOUNTADMIN

D. ROOT

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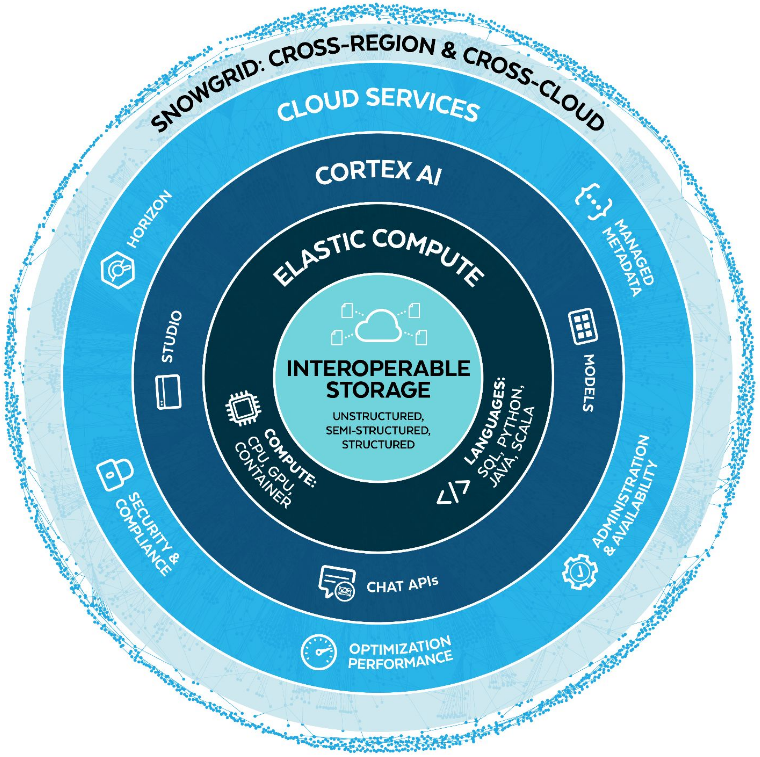
Summary for each Domain from outline

1. Snowflake AI Data Cloud Features and Architecture

Snowflake Data Cloud Features and Architecture

Snowflake Architecture



Snowflake Data Cloud Features and Architecture

Storage Layer

- Hybrid-Columnar storage (sets of rows stored in columns)
- Automatic micro-partitioning
- Native Semi-Structured data support and optimization

MICRO-PARTITIONS

- Contiguous units of storage that hold table data
- Generally 10MB to 16MB (Compressed)
- Many micro-partitions per table
- IMMUTABLE !
- Services layer stores metadata about every micro-partition
  - MIN/MAX (Range of values in each column)
  - Number of distinct values
  - NULL count

Snowflake Data Cloud Features and Architecture

Compute Layer

- Clusters of compute resources with CPU, memory, and disk
- Compute resources double with each size increase
- Jobs requiring compute run on virtual warehouses
- While running, a virtual warehouse consumes Snowflake credits

WAREHOUSE TYPE

- **Standard**
  - Always a single cluster
  - Scale up to a larger virtual warehouse for more compute
- **Multi-Cluster Warehouse (MCW)**
  - One or more clusters
  - Automatically scaled by Snowflake
  - Enterprise Edition feature

Snowflake Data Cloud Features and Architecture

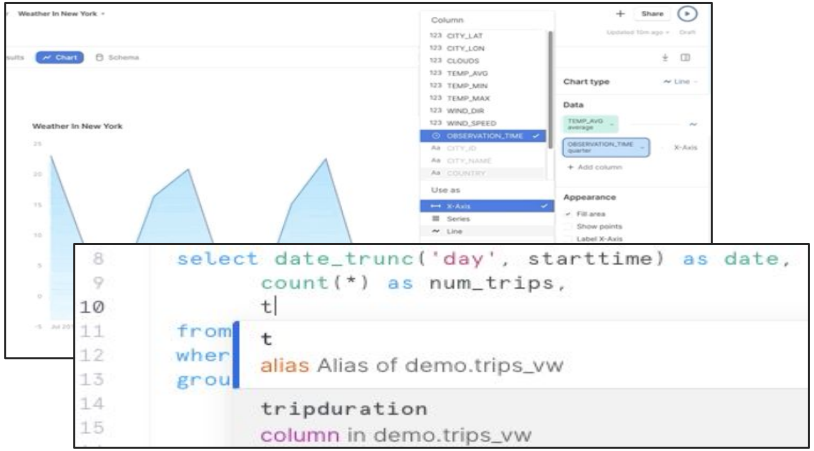
Cloud Service Layer

- MANAGEMENT
- QUERY HANDLING
- SECURITY
- METADATA MANAGEMENT
- SERVERLESS FEATURES COMPUTE
  - Clustering
  - Snowpipe
  - Materialized view maintenance
  - Replication
  - Search optimizations

- Short amounts of compute used for objects and queries
  - Cache lookups
  - Query optimizer
  - Object creation or deletion
- Customers are charged for cloud services compute that exceeds 10% of the total daily compute used from all sources
- Provided by the cloud services layer, not the compute layer

Snowflake Data Cloud Features and Architecture

Snowsight



- Web-based UI
- Perform Snowflake operations, including:
  - Account management
  - Loading data
  - Querying data
  - Building visualizations and dashboards
  - Managing Snowflake objects
  - Sharing data
  - Monitoring account activities

Snowflake Data Cloud Features and Architecture

Snowflake Driver / Connecters / Connection Options

DRIVER SUPPORT

- GO
- JDBC
- Microsoft .NET
- Node.js
- ODBC
- PHP PDO

CONNECTORS

- Snowflake connector for Python
- Snowflake connector for Spark
- Snowflake connector for Kafka

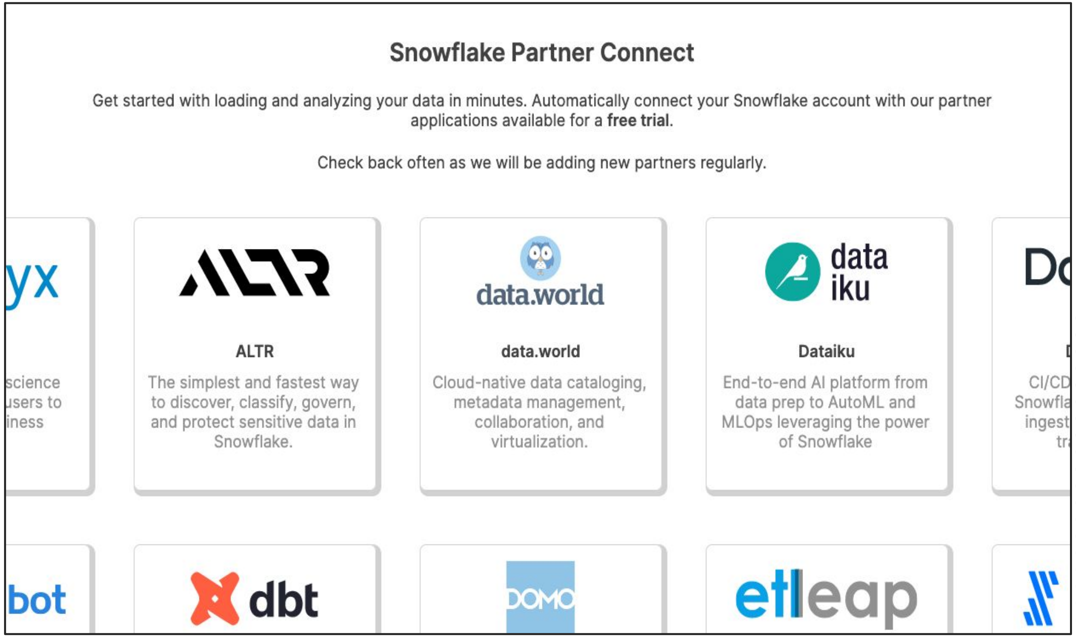
CONNECTION OPTIONS

- Snowsight
- SnowSQL
- Third-Party Solutions

Snowflake Data Cloud Features and Architecture

Partner Connect

- Create trial account with selected Snowflake Partners
- Use limited to ACCOUNTADMIN



Snowflake Data Cloud Features and Architecture

Data Types

Data Types	Data Types in Snowflake
NUMBER, DECIMAL, NUMERIC	NUMBER(38,0) – unless you specify a different scale and precision
INT, INTEGER, BIGINT, SMALLINT	NUMBER – cannot specify scale or precision
DOUBLE, DOUBLE PRECISION, FLOAT, FLOAT4, FLOAT8, REAL	DOUBLE
VARCHAR, STRING, TEXT	VARCHAR – max size is 16,777,216 bytes
CHAR, CHARACTER	VARCHAR(1) – though you can specify a different length
BINARY, VARBINARY	BINARY
BOOLEAN	BOOLEAN

Snowflake Data Cloud Features and Architecture

Account Objects / Database and Schema Objects

Account Objects

- Users
- Roles
- Virtual warehouses
- Resource monitors
- Integrations
- Databases

Database / Schema Objects

- An account has many databases, used to logically organize data
- A database contains schemas
  - Each database belongs to a single account
  - Each schema belongs to a single database
- A Schema is a logical grouping of objects
  - Tables / Views / Functions and Stored Procedures / Etcs.

Snowflake Data Cloud Features and Architecture

Table / View Types

Table Type

- Permanent Table (Fail-safe/Time Travel)
- Transient Table (Time Travel)
- Temporary Table (Time Travel)
- External Table

View Type

- Standard View
- Secure View (Share)
- Materialized View
- Secure Materialized View(Share)

Snowflake Data Cloud Features and Architecture

UDF / UDTF / Stored Procedures

User-Defined Functions

- Can be written in
  - JavaScript / SQL
  - Scala or Java (using Snowpark)
- Return scalar value or a set of row(table function)

User-Defined Table Functions

- Return as table
- Can be in From clause.

Stored Procedures

- DDL / DML permitted / may return a value
- Support Transaction inside procedure
- Can be written in JavaScript / Snowflake Scripting and Java / Python / Scalar for using Snowpark API
- Delimiter for Stored Procedure : $$ or ‘’

Snowflake Data Cloud Features and Architecture

Other Objects

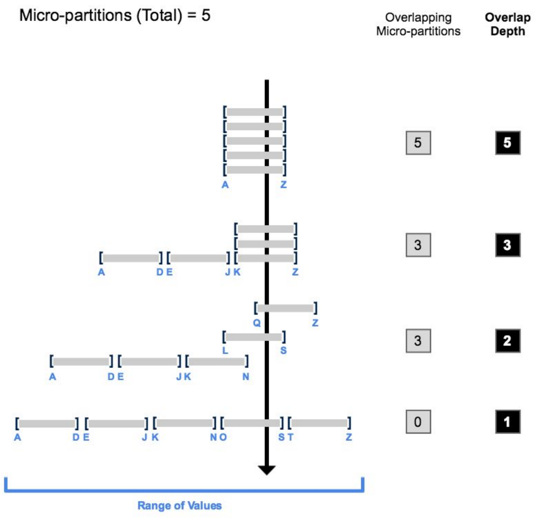
Other Objects

- Streams
- Tasks
- Pipes
- Shares
- Sequence

Snowflake Data Cloud Features and Architecture

Clustering

Clustering Depth



Monitoring Clustering Information

- SYSTEM$CLUSTERING_DEPTH
- SYSTEM$CLUSTERING_INFORMATION (including clustering depth)

Add/Drop Clustering Key

- Add Clustering key

```
ALTER TABLE <name> CLUSTER BY ( <expr1> [ , <expr2> ... ] )
```

- Drop Clustering Key

```
ALTER TABLE <name> DROP CLUSTERING KEY
```

Snowflake Data Cloud Features and Architecture

Data Storage Monitoring

Monitoring Storage Costs

- Web UI : **Admin > Usage > Storage**
- Query TABLE_STORAGE_METRICS (in ACCOUNT_USAGE or INFORMATION_SCHEMA)]
- Query DATABASE_STORAGE_USAGE_HISTORY (in ACCOUNT_USAGE)

Snowflake Data Cloud Features and Architecture

Search Optimization

Search Optimization Service

- Equality searches.
- Substring and regular expression searches.
- Searches of fields in VARIANT.
- Searches of GEOGRAPHY columns using geospatial functions
- Require Privileges : OWNERSHIP / ADD SEARCH OPTIMIZATION

Quiz

1. **What is the recommended Snowake data type to store semi-structured data like JSON?**

A. VARCHAR

B. RAW

C. LOB

D. VARIANT

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2. What is the lowest Snowake edition that offers Time Travel up to 90 days?

A. Standard Edition

B. Premier Edition

C. Enterprise Edition

D. Business Critical Edition

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2. Account Access and Security

Account Access and Security

Network Policy

Configure Network Policy

- User must be ACCOUNTADMIN or SECURITYADMIN
- ALLOWED_IP_LIST is required, BLOCKED_IP_LIST is optional

Set Network Policy

- Only one network policy can be active for an account or user
- The BLOCKED_IP_LIST is applied before the ALLOWED_IP_LIST

Bypassing Network Policy

- possible to temporarily bypass a network policy for a set number of minutes by configuring the user object property MINS_TO_BYPASS_NETWORK_POLICY
- Only Snowflake can set the value for this object property. contact Snowflake Support

Account Access and Security

Access Control

- Combination of Role-Based (RBAC) and Discretionary Control
  - Role-based: Access privileges are assigned to roles
  - Discretionary: Each object has an owner, which can grant others access to that object
- Privileges that can be set depend on the securable object
- Privileges can be broad or granular
- GRANT / REVOKE

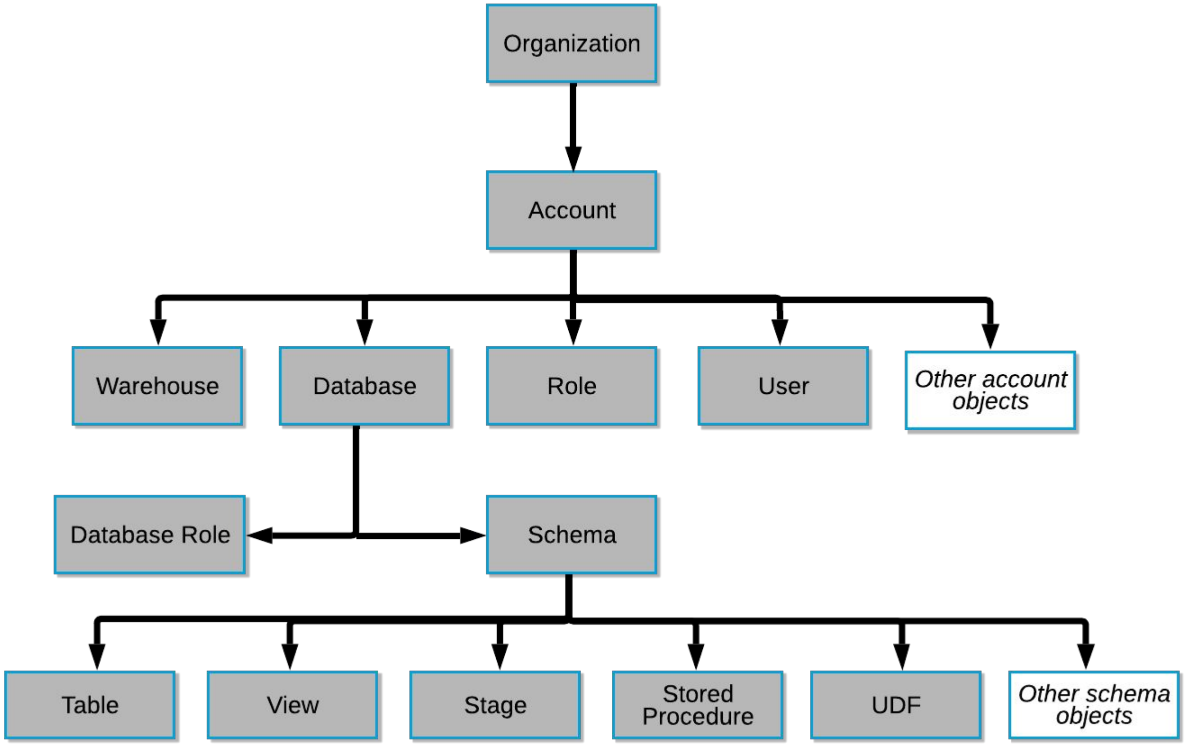
Account Access and Security

Authentication

- Username and password
- Multi-factor authentication
- Federated authentication (SAML 2.0)
- Other authentication methods (OAuth, Key-Pair)

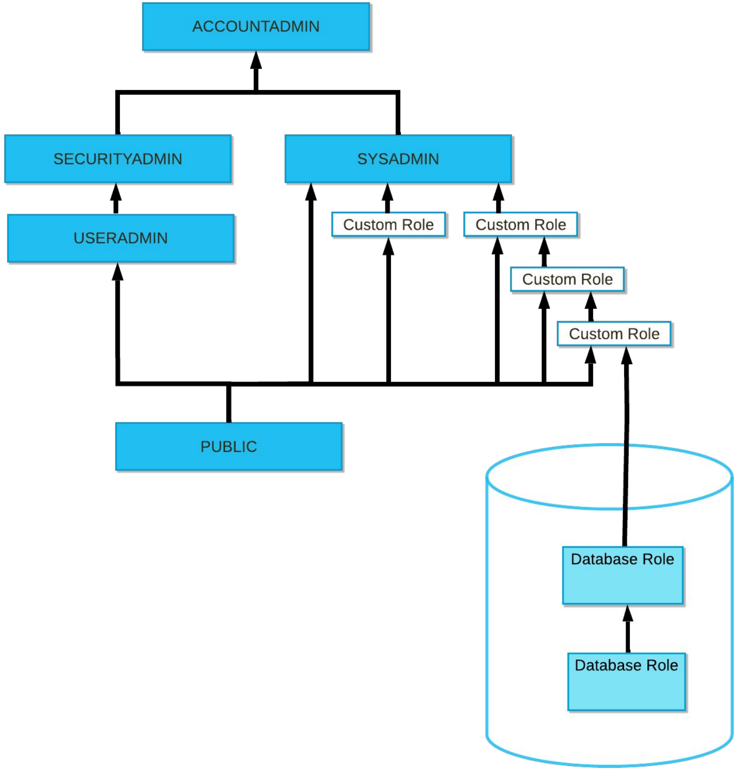
Account Access and Security

Object hierarchy



Account Access and Security

Role Hierarchy and inheritance



Account Access and Security

Organizations

Organizations

- An organization is a first-class Snowflake object that links the accounts owned by your business entity

ORGADMIN Role

- Create an account in the organization
- View / show all accounts within organization and View / show a list of regions enabled for the organization
- View usage information for all accounts in the organization
- Enable database replication for an account in the organization
- If wants to delete account in the organization, contact snowflake support

Account Access and Security

Secure Views / UDFs

- Prevent data from being indirectly exposed through programmatic methods
- Only authorized users can see the view or UDF definition
- Secure view and secure UDFs bypass some query optimizations to achieve greater security

Account Access and Security

Row and Column-Level Protection

Protect row-level access

- Row access policies
- Context function (such as CURRENT_USER, CURRENT_ROLE, etc.)
- Secure Views

Protect column-level access

- Dynamic data masking
- External tokenization
- Secure views

Account Access and Security

Information Schemas

- Built-in, read-only schema
- Exists for each database
- System-defined views and table functions
    - Views for all objects in the database
    - Views for account-level objects
    - Table functions for historical and usage data
- Provide extensive "logging" metadata information

Some Views in Information_schema

- APPLICABLE_ROLES
- COLUMNS
- DATABASES
- ENABLED_ROLES
- FILE FORMATS
- FUNCTIONS
- LOAD_HISTORY
- PIPES
- PROCEDURES
- SEQUENCES
- STAGES
- TABLE_PRIVILEGES
- TABLE_STORAGE_METRICS
- TABLES
- USAGE_PRIVILEGES
- VIEWS

Table Functions in Information_schema

- AUTOMATIC_CLUSTERING_HISTORY
- DATABASE_STORAGE_USAGE_HISTORY
- LOGIN_HISTORY
- LOGIN_HISTORY_BY_USER
- MATERIALIZED_VIEW_REFRESH_HISTORY
- QUERY_HISTORY
- WAREHOUSE_LOAD_HISTORY

Account Access and Security

Snowflake Database / Access History

Snowflake Database

- Shared by Snowflake as a collection of secured views.
- By default, only ACCOUNTADMIN can access the SNOWFLAKE database and schemas
- Privileges can be granted to other roles

  GRANT IMPORTED PRIVILEGES ON DATABASE SNOWFLAKE TO ROLE <role>;

Access History

- Can access SNOWFLAKE.ACCOUNT_USAGE.LOGIN_HISTORY view for specific user's login history.

Account Access and Security

Account_usage VS. Information_schema

ACCOUNT_USAGE (IN SNOWFLAKE DATABASE)	INFORMATION_SCHEMA (IN INDIVIDUAL DATABASES)
Includes dropped objects	Does not include dropped objects
45 minutes to 3 hours latency (varies by view)	Low latency
Data retained for 1 year	Data retained up to 6 months (varies by view/table function)

Quiz

1. **Which is true of Snowake network policies? A Snowake network policy: (Choose two.)**

A. Is available to all Snowake Editions

B. Is only available to customers with Business Critical Edition

C. Restricts or enables access to specific IP addresses

D. Is activated using an ALTER DATABASE command

1. **Which is true of Snowake network policies? A Snowake network policy: (Choose two.)**

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   D. Is activated using an ALTER DATABASE command

2. Where would a Snowflake user and information about query activity from 90 days ago?

A. account_usage.query_history view

B. account_usage.query_history_archive view

C. information_schema.query_history view

D. information_schema.query_history_by_session view

2. Where would a Snowflake user and information about query activity from 90 days ago?

A. account_usage.query_history view

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C. information_schema.query_history view

D. information_schema.query_history_by_session view

3. Performance Concepts

Performance Concepts

Explain

```
EXPLAIN [ USING { TABULAR | JSON | TEXT } ] <statement>
```

- Displays the logical execution steps without executing
- Results show in JSON, text or tabular format
- Helpful for evaluating micro-partition pruning and joins

What's In a PLAN?

- Partition Pruning
- Join Ordering
- Join Types

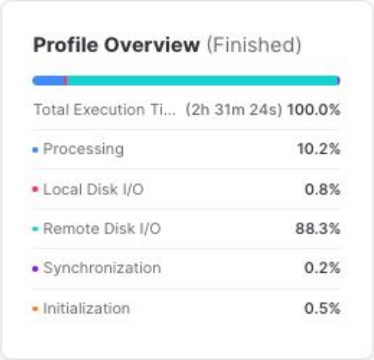
Performance Concepts

Query History / Profile

Query History

- **Activity > Query History**
- Click Query ID to access Query Profile / Details

What's in Profile Overview?



- Processing: query processing
- Local Disk I/O: communication within the cluster
- Remote Disk I/O: communication between storage and compute layers
- Synchronization: managing activities between processes
- Initialization: includes virtual warehouse startup time

Performance Concepts

Statistics

What's the important things in Statistics?

Statistics

Scan progress	100.00%
Bytes scanned	1.13TB
Percentage scanned from cache	0.00%
Bytes written to result	20.29GB
Partitions scanned	721507
Partitions total	721507
Bytes spilled to local storage	373.89GB
Bytes spilled to remote storage	105.81GB

- Percentage scanned from cache
  - Percentage of the required data that was already on the warehouse cache
- Partitions scanned vs Partitions total
  - Indication of how much data the optimizer was able to prune
- Bytes spilled to local storage
  - Insufficient memory
- Bytes spilled to remote storage
  - Grossly insufficient memory

Performance Concepts

Warehouse Size / Compute Credits

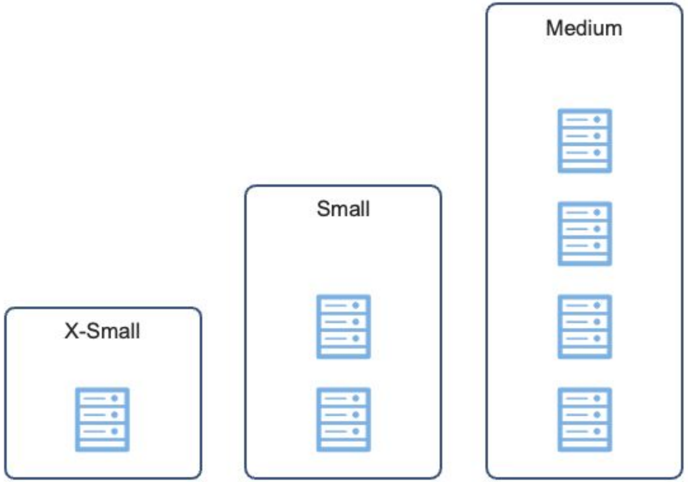
Warehouse Size	Credits / Hour	Credits / Second
XS	1	0.0003
S	2	0.0006
M	4	0.0011
L	8	0.0022
XL	16	0.0044
2XL	32	0.0089
3XL	64	0.0178
4XL	128	0.0356
5XL	256	0.0711
6XL	512	0.1422

- How you are billed for compute usage
- Charged by the number of running clusters, their sizes, and how long they are running
- Compute is charged per-second, with a one-minute minimum
- 5XL and 6XL virtual warehouses are currently in preview
- METERING_HISTORY View in Account_Usage schema can used to return the hourly credit usage

Performance Concepts

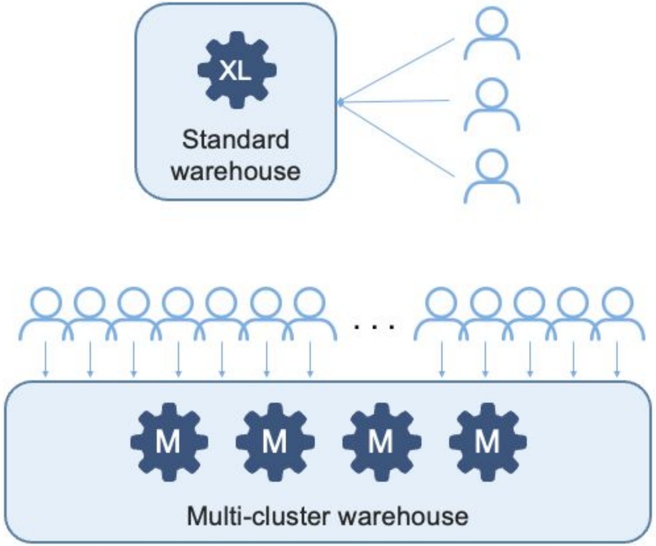
Warehouse (SCALE UP / SCALE OUT)

SCALE UP (OR DOWN)



SCALE OUT (OR BACK)

HANDLE GREATER CONCURRENCY



Performance Concepts

Warehouse (Multi-cluster Warehouse)

- Maximum number of clusters, greater than 1 (up to 10).
- Minimum number of clusters, equal to or less than the maximum (up to 10).

Maximized vs. Auto-Scale

- Maximized : same value for both maximum and minimum number of clusters and must be larger than 1
- Auto-Scale : specifying different values for maximum and minimum number of clusters

Performance Concepts

Warehouse (Multi-Clustering Policy)

Standard

- Minimizes queuing and starts additional clusters
- Cluster Starts: Immediately when either a query is queued or the system detects that there's one more query than the currently-running clusters can execute.
- Cluster Shuts down : After 2-3 consecutive checks (performed at 1 minute intervals) determine that the load on the least-loaded cluster could be redistributed to the others without spinning up the cluster again.

Economy

- Favors keeping running clusters fully-loaded rather than starting additional clusters; could result in jobs being queued rather than starting additional Clusters.
- Cluster Starts **: Only** if the system estimates there's enough query load to keep the cluster busy for at least 6 minutes
- Cluster Shuts down : After 5-6 consecutive checks (performed at 1 minute intervals), determine that the load on the least-loaded cluster could be redistributed to the others without spinning up the cluster again.

Performance Concepts

Warehouse Setting / Access

Options for Creating or Altering Warehouse

```
WAREHOUSE_TYPE = STANDARD | SNOWPARK-OPTIMIZED
WAREHOUSE_SIZE = XSMALL | SMALL | MEDIUM | LARGE | XLARGE | XXLARGE | XXXLARGE | X4LARGE |
WAIT_FOR_COMPLETION = TRUE | FALSE
MAX_CLUSTER_COUNT = <num>
MIN_CLUSTER_COUNT = <num>
SCALING_POLICY = STANDARD | ECONOMY
AUTO_SUSPEND = <num>
AUTO_RESUME = TRUE | FALSE
RESOURCE_MONITOR = <monitor_name>
COMMENT = '<string_literal>'
ENABLE_QUERY_ACCELERATION = TRUE | FALSE
QUERY_ACCELERATION_MAX_SCALE_FACTOR = <num>
```

Privileges for Warehouse

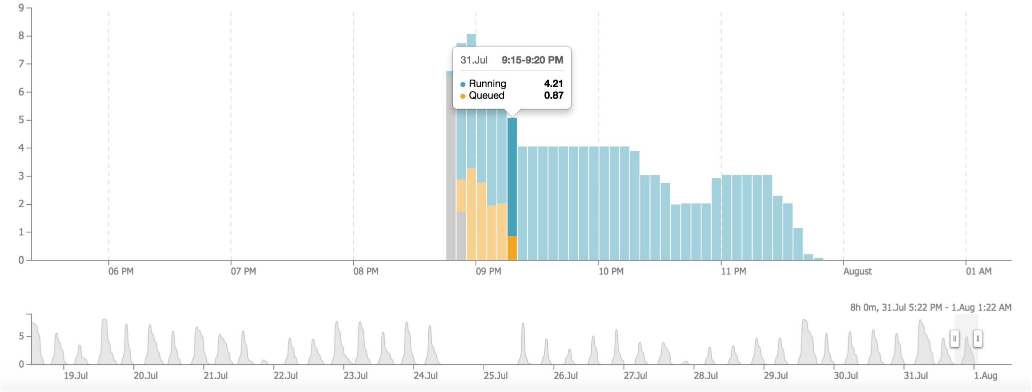
- USAGE / MODIFY / OWNERSHIP / MONITOR / OPERATE / ALL

Performance Concepts

Monitoring warehouse loads

Viewing the Load Monitoring Chart

- Classic Console > Warehouse > Warehouse_name > Warehouse Load Over Time



* https://docs.snowflake.com/en/user-guide/warehouses-load-monitoring

Performance Concepts

Resource Monitors

- ACCOUNTADMIN privileges are required to set up Resource Monitors
- Resource Monitor attributes:
  - Credit quota
  - Level (account, warehouse, or warehouses)
  - Schedule (start, end, frequency)
  - Action (notify, suspend, suspend immediate)
- Not intended to strictly control usage to the second
  - Quota may be exceeded before action is triggered
- Must enable notifications in Preferences

Performance Concepts

Materialized Views

- Query results contain a small number of rows and/or columns relative to the base table
- Query results contain results that require significant processing:
  - Analysis of semi-structured
  - Aggregates that take a long time to calculate
- The query is on an external table
- The view's base table does not change frequently
- Materialized views can improve the performance of queries that use the same subquery results repeatedly
- Materialized views are automatically and transparently maintained by Snowflake

Quiz

1. Several users are using the same virtual warehouse. The users report that the queries are running slowly, and that many queries are being queued. What is the recommended way to resolve this issue?

A. Reduce the warehouse STATEMENT_QUEUED_TIMEOUT_IN SECONDS parameter.

B. Reduce the warehouse AUTO_SUSPEND parameter.

C. Increase the warehouse MAX_CONCURRENCY_LIMIT parameter.

D. Increase the warehouse MAX_CLUSTER_COUNT parameter.

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B. Reduce the warehouse AUTO_SUSPEND parameter.

C. Increase the warehouse MAX_CONCURRENCY_LIMIT parameter.

D. Increase the warehouse MAX_CLUSTER_COUNT parameter.

2. When reviewing a query profile, what is a symptom that a query is too large to fit into the memory?

A. A single join node uses more than 50% of the query time

B. Partitions scanned is equal to partitions total

C. An AggregateOperator node is present

D. The query is spilling to remote storage

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4. Data Loading and Unloading

Data Loading and Unloading

Stages and Stage types

Stages

- Snowflake refers to the location of data files in cloud storage as a stage

Stage types

- External Stages : Amazon S3, Google Cloud Storage, Microsoft Azure
- Internal Stages(Reference) : User (@~), Table(@%), Named(@)

Snowpipe Auto-ingest : only external stage
REST api ingest : Internal and external stage

Data Loading and Unloading

File size / File format / Folder Structures for Loading data

File Size

- Recommended : 100MB to 250MB(compressed)
- Split large files to before loading

File format

- CSV, JSON, Avro, ORC, Parquet, XML for Data Loading
- Order of precedence : 1. COPY INTO TABLE / 2. Stage definition / 3. Table definition

Folder Structure

- Organize data in logical paths (e.g., subject area and create date)
  ex. /Marketing/daily/2023/03/29/

Data Loading and Unloading

Commands for Loading Data

- CREATE PIPE
- COPY INTO
- GET / PUT (loading data to Internal Stages)
- INSERT / INSERT OVERWRITE
- STREAM
- TASK
- VALIDATE (COPY INTO option : RETURN_<n>_ROWS | RETURN_ERRORS | RETURN_ALL_ERRORS)

Data Loading and Unloading

File formats for unloading / Empty strings and NULL handing for unloading

File format

- CSV, JSON, Parquet for Data Unloading
- Specifying File Format Options
    - COPY INTO <location>
    - Named stage definition
    - Table definition

Empty Strings / NULL

- COPY INTO <location> options
    - NULL_IF
    - EMPTY_FIELD_AS_NULL

Data Loading and Unloading

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Empty Strings / NULL

- COPY INTO <location> options
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Data Loading and Unloading

Unloading a single file / relational tables

Unloading a single file

```
copy into @mystage/myfile.csv.gz from mytable
file_format = (type=csv compression='gzip')
single=true
max_file_size=4900000000;
```

Unloading a Relational Table

```
COPY INTO @mystage/myfile.parquet FROM (SELECT id, name, start_date FROM mytable)
  FILE_FORMAT=(TYPE='parquet')
  HEADER = TRUE;

COPY INTO @mystage
 FROM (SELECT OBJECT_CONSTRUCT('id', id, 'first_name', first_name, 'last_name', last_name, '
 FILE_FORMAT = (TYPE = JSON);
```

Data Loading and Unloading

Commands for Unloading Data

- LIST (list files in external stages or internal stages)
- COPY INTO
- CREATE FILE FORMAT
- CREATE FILE FORMAT ... CLONE
- ALTER FILE FORMAT
- DROP FILE FORMAT
- DESCRIBE FILE FORMAT
- SHOW FILE FORMAT

Quiz

1. **Select the different types of Internal Stages: (Choose three.)**

A. Named Stage

B. User Stage

C. Table Stage

D. Schema Stage

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A. Named Stage

B. User Stage

C. Table Stage

D. Schema Stage

2. Which of the following statements would be used to export/unload data from Snowake?

A. COPY INTO @stage

B. EXPORT TO @stage

C. INSERT INTO @stage

D. EXPORT_TO_STAGE(stage => @stage, select => 'select * from t1');

2. Which of the following statements would be used to export/unload data from Snowake?

A. COPY INTO @stage

B. EXPORT TO @stage

C. INSERT INTO @stage

D. EXPORT_TO_STAGE(stage => @stage, select => 'select * from t1');

5. Data Transformations

Data Transformations

Work with standard data

Estimating functions

- APPROXIMATION
  ex) APPROX_COUNT_DISTINCT
- FREQUENCY ESTIMATION
  ex) APPROX_TOP_K
- SIMILARITY ESTIMATION
  ex) APPROXIMATE_SIMILARITY

- Small margin of error
- Uses a constant amount of memory regardless of the size of the input

Data Transformations

SAMPLING

Sampling Examples

- SELECT * FROM table SAMPLE (10);
- SELECT * FROM table SAMPLE BLOCK (10);
- SELECT * FROM table SAMPLE (5) SEED (4444);
- SELECT * FROM table SAMPLE (10000 rows);
- SELECT AVG(price) FROM (SELECT price FROM table SAMPLE (5));

Data Transformations

Supported Function Types

Function Types

- Scalar Functions
- Aggregate Functions
- Window Functions
- Table Functions
- System Functions
- User-defined Functions
- External Functions

Data Transformations

Semi-structured data

- Snowflake is storing semi-structured data as Variant Data type
- Variant contains nested ARRAYs and OBJECTs
- File Format : JSON, Avro, ORC, Parquet, XML
- Data Size Limitation : 16MB individual rows

Data Transformations

Querying Semi-structured data

- Insert a colon : between the VARIANT column name and any first-level element: <column>:<level1_element>.

```
SELECT src:salesperson.name
    FROM car_sales
    ORDER BY 1;
+----------------------+
| SRC:SALESPERSON.NAME |
|----------------------|
| "Frank Beasley"      |
| "Greg Northrup"      |
+----------------------+
```

Dot Notation

```
SELECT src['salesperson']['name']
    FROM car_sales
    ORDER BY 1;
+----------------------------+
| SRC['SALESPERSON']['NAME'] |
|----------------------------|
| "Frank Beasley"            |
| "Greg Northrup"            |
+----------------------------+
```

Bracket Notation

- column name is case-insensitive but element names are case-sensitive

  - src:salesperson.name
  - SRC:salesperson.name
  - SRC:Salesperson.Name

Data Transformations

Casting Values in Semi-structured data / FLATTEN Function to Parse Arrays

- Casting Values by using ':'

```
SELECT src:vehicle[0].price::NUMBER * 0.10 AS tax
    FROM car_sales
    ORDER BY tax;
+--------+
|    TAX |
|--------|
| 2027.5 |
| 2350.0 |
+--------+
```

- FLATTEN() extracts data from nested elements

```
SELECT
  value:name::string as "Customer Name",
  value:address::string as "Address"
  FROM
    car_sales
  , LATERAL FLATTEN(INPUT => SRC:customer);

+--------------------+-------------------+
| Customer Name      | Address           |
|--------------------+-------------------|
| Joyce Ridgely      | San Francisco, CA |
| Bradley Greenbloom | New York, NY      |
+--------------------+-------------------+
```

Data Transformations

FLATTEN function output

```
+-----+-------+-------+--------+--------+-------+
| SEQ |   KEY | PATH | INDEX | VALUE | THIS |
|-----+-------+-------+--------+--------+-------|
```

Copy

SEQ A unique sequence number associated with the input record; the sequence is not guaranteed to be gap-free or ordered in any particular way.

KEY For maps or objects, this column contains the key to the exploded value.

PATH The path to the element within a data structure which needs to be flattened.

INDEX The index of the element, if it is an array; otherwise NULL.

VALUE The value of the element of the flattened array/object.

THIS The element being flattened (useful in recursive flattening).

Data Transformations

Unstructured data (Directory Table / Types of URLs to Access Files)

Directory Table

- Directory tables are similar to external tables in that they store file-level metadata about the data files in a stage. Query a directory table to retrieve the Snowflake-hosted file URL to each file in the stage.

Types of URLs to Access Files

- Scoped URL
- File URL
- Pre-signed URL

Column	Data Type	Description
RELATIVE_PATH	TEXT	Path to the files to access using the file URL.
SIZE	NUMBER	Size of the file (in bytes).
LAST_MODIFIED	TIMESTAMP_LTZ	Timestamp when the file was last updated in the stage.
MD5	HEX	MD5 checksum for the file.
ETAG	HEX	ETag header for the file.
FILE_URL	TEXT	Snowflake-hosted file URL to the file. The file URL has the following format:

Data Transformations

Unstructured data (SQL Functions)

SQL Function	Description
GET_STAGE_LOCATION	Returns the URL for an external or internal named stage using the stage name as the input.
GET_RELATIVE_PATH	Extracts the path of a staged file relative to its location in the stage using the stage name and absolute file path in cloud storage as inputs.
GET_ABSOLUTE_PATH	Returns the absolute path of a staged file using the stage name and path of the file relative to its location in the stage as inputs.
GET_PRESIGNED_URL	Generates the pre-signed URL to a staged file using the stage name and relative file path as inputs. Access files in an external stage using the function.
BUILD_SCOPED_FILE_URL	Generates a scoped Snowflake-hosted URL to a staged file using the stage name and relative file path as inputs.
BUILD_STAGE_FILE_URL	Generates a Snowflake-hosted file URL to a staged file using the stage name and relative file path as inputs.

Quiz

1. **Which URL type allows users to access unstructured data without authenticating into Snowake or passing an authorization token?**

   A. Pre-signed URL

   B. Scoped URL

   C. Signed URL

   D. File URL

1. **Which URL type allows users to access unstructured data without authenticating into Snowake or passing an authorization token?**

   A. **Pre-signed URL**

   B. Scoped URL

   C. Signed URL

   D. File URL

1. **What will be the output of the below query against the table name gold_data?**
   select * from gold_data tablesample (100);

   A. It will return an empty sample.
   B. It will return a random 100 rows.
   C. It will return an entire table.
   D. It will produce an error message.

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6. Data Protection and Data Sharing

Data Protection and Data Sharing

Time Travel

- Access historical data at any point within a defined retention period
- UNDO common mistakes
- Protect against accidental or intentional modification, removal, or corruption
- Fix drops, deletes, edits
- Backup/Restore from time or ID
- Time interval depends on value of DATA_RETENTION_TIME_IN_DAYS
- Default Time Travel : 1 Day , *Enterprise or above up to 90 days

Data Protection and Data Sharing

Fail-Safe

- Non-configurable, 7-day retention for historical data after Time Travel expiration
- Only accessible by Snowflake personnel
- Admins can view Fail-safe use in the Snowflake Web UI under Account > Billing and Usage
- Table_storage_metrics in Information_schema/account_usage

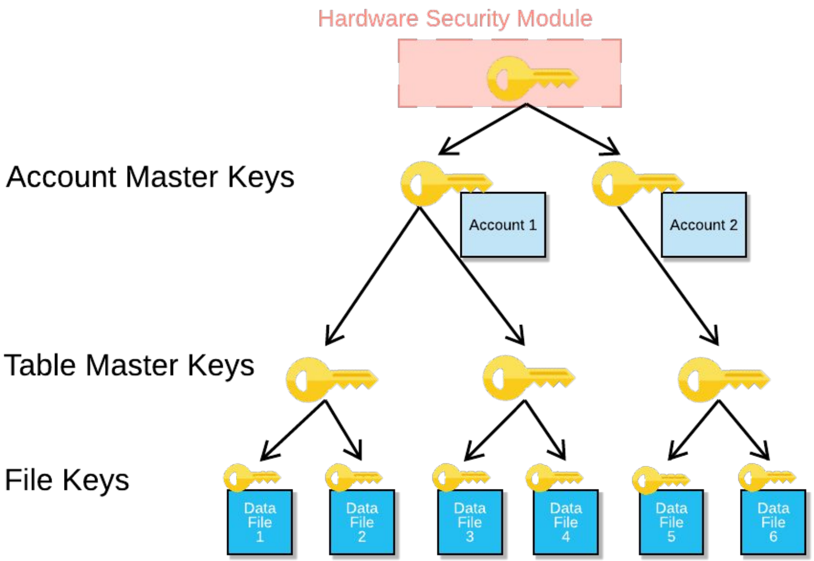
Data Protection and Data Sharing

Cloning

- Quickly take a “snapshot” of any table, schema, or database
    - When the clone is created, micro-partitions are fully shared
- No additional storage costs until changes are made to the original or the clone
- Often used to quickly spin up Dev or Test environments
- Effective “backup” option as well

Data Protection and Data Sharing

Data Encryption



- Encryption at rest and in motion
- Hierarchical key model
  - Root key
  - Account master keys
  - Table master keys
  - File (micro-partition) keys
- Automatic key rotation every 30 days
- Optional periodic rekeying (Enterprise edition and above)
- Bring Your Own Key (BYOK)

Data Protection and Data Sharing

Replication

- Available on all editions of Snowflake
- Keeps data synchronized between one or more accounts within the same organization
- Unit of replication is a database (permanent or transient)
- Secondary database is read-only
- Source and destination accounts must be in the same organization
- Database replication currently excludes these database objects:
  - Temporary tables, temporary stages, and external tables
  - Pipes, streams, and tasks
- Privileges on database objects are not replicated to a secondary database
- Enable client redirect to maintain client connections after failover (Business Critical feature)

Data Protection and Data Sharing

Data Sharing

- Direct Data Sharing
- Snowflake Data Exchange
- Snowflake Marketplace
- Sharing objects
  - Tables
  - External tables
  - Secure views
  - Secure materialized views
  - Secure UDFs

Quiz

1. **Which of the following Snowflake objects can be shared using a secure share? (Choose two.)**

   A. Materialized views

   B. Sequences

   C. Procedures

   D. Tables

   E. Secure User Defined Functions (UDFs)

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   A. Materialized views

   B. Sequences

   C. Procedures

   D. Tables

   E. Secure User Defined Functions (UDFs)

2. Fail-safe is unavailable on which table types? (Choose two.)

A. Temporary

B. Transient

C. Provisional

D. Permanent

2. **Fail-safe is unavailable on which table types? (Choose two.)**

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Sharing by Peers

THANK YOU

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